



6362 - Breaking the degeneracy: substellar anchors for evolutionary models

Cycle: 3, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	HD13724	NIRSpec IFU Spectroscopy	(1) HD-13724
	2	HIP99770	NIRSpec IFU Spectroscopy	(2) HIP-99770
	3	GJ758	NIRSpec IFU Spectroscopy	(13) GJ-758-B

Folder	Observation	Label	Observing Template	Science Target
	4	HD33632	NIRSpec IFU Spectroscopy	(4) HD-33632
	5	HD47127	NIRSpec IFU Spectroscopy	(11) HD-47127-B
	6	HIP21152	NIRSpec IFU Spectroscopy	(6) HIP-21152
	7	HD72946	NIRSpec IFU Spectroscopy	(7) HD-72946
	8	HR7672	NIRSpec IFU Spectroscopy	(12) HR-7672-B
	9	HD4747	NIRSpec IFU Spectroscopy	(9) HD-4747
	10	HD984	NIRSpec IFU Spectroscopy	(10) HD-984
	11	HIP-99770-2	NIRSpec IFU Spectroscopy	(2) HIP-99770

ABSTRACT

We propose to build the gold standard of calibrators for brown dwarf and exoplanet models with JWST. By definition, brown dwarfs have insufficient masses to sustain hydrogen fusion in their cores, which causes them to cool down over time since they never reach thermodynamical equilibrium. This age-mass-luminosity degeneracy is the principal challenge to the characterization of brown dwarfs and exoplanets alike. Brown dwarf evolutionary and spectral models are routinely used to compare with exoplanet data, which intrinsically suffer from lower SNR due to their proximity to their host star. In order to anchor evolutionary models in measured physical parameters and chemical abundances, we request NIRSpec/IFU spectroscopy on a sample of brown dwarf companions to solar-type stars with measured dynamical masses from previous direct imaging in age-calibrated benchmark systems. This carefully mass- and age-calibrated sample will set the basis for testing, calibrating, and identifying physical processes missing in evolutionary and spectral models.

OBSERVING DESCRIPTION

Our goal is to obtain F290LP/G395H spectra of a sample of directly-imaged brown dwarf companions to solar-type stars.

All observations have been simulated with the ETC, on notebook #177465.

The spatially-resolved IFU cubes will allow modeling and subtraction of the stellar PSF halo to obtain high SNR spectra. The higher contrast companions will utilize the observing and data analyses strategy following Ruffio et al. 2023. We will use the 9-pt small size CYCLING dithering pattern to improve the spatial sampling for all IFU obs. Detector settings are chosen to ensure uniform SNR of all the companions while also minimizing saturation of the host star, which often requires using as few groups as possible.

Dithers

As recommended, for good spatial sampling of the PSF halo, in each observation we use the CYCLING dither, size small, with 9 points to prevent the companions from being too close to the edges.

Offset Pointing Special Requirements

We apply an offset for each of our targets. This places the companion as central in the FOV as possible and at least a few hundreds of milliarcsecs away from the edge of the FOV, providing good tolerance to pointing uncertainties, while keeping the host PSF in a corner of the field of view.

PA special requirement

PA range is required to avoid placing the science target along the spiders.

Proposal 6362 - Targets - Breaking the degeneracy: substellar anchors for evolutionary models

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	HD-13724	RA: 02 12 20.6773 (33.0861554d) Dec: -46 48 58.95 (-46.81637d) Equinox: J2000	Proper Motion RA: -30.624 mas/yr Proper Motion Dec: -69.08799998655013 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[G stars] Extended=NO				
	(2)	HIP-99770	RA: 20 14 32.0324 (303.6334683d) Dec: +36 48 22.70 (36.80631d) Equinox: J2000	Proper Motion RA: 68.09 mas/yr Proper Motion Dec: 69.404 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[A stars] Extended=NO				
	(3)	GJ-758	RA: 19 23 34.0132 (290.8917217d) Dec: +33 13 19.08 (33.22197d) Equinox: J2000	Proper Motion RA: 81.966 mas/yr Proper Motion Dec: 160.158 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[K stars] Extended=NO				
	(4)	HD-33632	RA: 05 13 17.4478 (78.3226992d) Dec: +37 20 14.32 (37.33731d) Equinox: J2000	Proper Motion RA: -144.922 mas/yr Proper Motion Dec: -136.77200008714863 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[F stars] Extended=NO				
	(5)	HD-47127	RA: 06 37 33.9005 (99.3912521d) Dec: +12 10 50.00 (12.18056d) Equinox: J2000	Proper Motion RA: -74.864 mas/yr Proper Motion Dec: -283.98199995081086 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[G stars] Extended=NO				
	(6)	HIP-21152	RA: 04 32 4.8086 (68.0200358d) Dec: +05 24 36.15 (5.41004d) Equinox: J2000	Proper Motion RA: 112.174 mas/yr Proper Motion Dec: 7.756 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[F stars] Extended=NO				
	(7)	HD-72946	RA: 08 35 51.2667 (128.9636113d) Dec: +06 37 21.95 (6.62276d) Equinox: J2000	Proper Motion RA: -136.593 mas/yr Proper Motion Dec: -137.14800006709993 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[G stars] Extended=NO				

Proposal 6362 - Targets - Breaking the degeneracy: substellar anchors for evolutionary models

(8)	HR-7672	RA: 20 04 6.2208 (301.0259200d) Dec: +17 04 12.68 (17.07019d) Equinox: J2000	Proper Motion RA: -387.472 mas/yr Proper Motion Dec: -419.4970000298781 mas/yr Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[G stars] Extended=NO			
(9)	HD-4747	RA: 00 49 26.7605 (12.3615021d) Dec: -23 12 44.87 (-23.21246d) Equinox: J2000	Proper Motion RA: 519.049 mas/yr Proper Motion Dec: 124.04099999999998 mas/yr Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[G stars] Extended=NO			
(10)	HD-984	RA: 00 14 10.2528 (3.5427200d) Dec: -07 11 56.81 (-7.19911d) Equinox: J2000	Proper Motion RA: 104.775 mas/yr Proper Motion Dec: -68.0160000001706 mas/yr Epoch of Position: 2000
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[F stars] Extended=NO			
(11)	HD-47127-B	RA: 06 37 33.9201 (99.3913338d) Dec: +12 10 51.88 (12.18108d) Equinox: J2000	Proper Motion RA: -74.864 mas/yr Proper Motion Dec: -283.98199995081086 mas/yr Epoch of Position: 2000
Comments: Category=Star Description=[Brown dwarfs, Exoplanet Systems] Extended=NO			
(12)	HR-7672-B	RA: 20 04 6.1899 (301.0257912d) Dec: +17 04 13.40 (17.07039d) Equinox: J2000	Proper Motion RA: -387.472 mas/yr Proper Motion Dec: -419.4970000298781 mas/yr Epoch of Position: 2000
Comments: Category=Star Description=[Brown dwarfs, Exoplanet Systems] Extended=NO			
(13)	GJ-758-B	RA: 19 23 33.9327 (290.8913862d) Dec: +33 13 18.31 (33.22175d) Equinox: J2000	Proper Motion RA: 81.966 mas/yr Proper Motion Dec: 160.158 mas/yr Epoch of Position: 2000
Comments: Category=Star Description=[Brown dwarfs, Exoplanet Systems] Extended=NO			

Proposal 6362 - Observation 1 - Breaking the degeneracy: substellar anchors for evolutionary models

Observation	Proposal 6362, Observation 1: HD13724											Tue Nov 25 17:00:27 GMT 2025
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	HD-13724	RA: 02 12 20.6773 (33.0861554d)				Proper Motion RA: -30.624 mas/yr					
			Dec: -46 48 58.95 (-46.81637d)				Proper Motion Dec: -69.08799998655013 mas/yr					
			Equinox: J2000				Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[G stars] Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	4	18	false	true	NONE	9	162	11817.001	177465
Special Requirements	Aperture PA Range 66.97164917 to 86.97164917 Degrees (V3 288.0 to 308.0) Offset -0.147 arcsec, 0.239 arcsec											

Proposal 6362 - Observation 2 - Breaking the degeneracy: substellar anchors for evolutionary models

Observation	Proposal 6362, Observation 2: HIP99770											Tue Nov 25 17:00:27 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(2)	HIP-99770	RA: 20 14 32.0324 (303.6334683d)				Proper Motion RA: 68.09 mas/yr					
	Dec: +36 48 22.70 (36.80631d)				Proper Motion Dec: 69.404 mas/yr							
	Equinox: J2000				Epoch of Position: 2000							
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
Category=Star												
Description=[A stars]												
Extended=NO												
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	5	6	false	true	NONE	9	54	4726.8	177465
Special Requirements	Aperture PA Range 296.97164917 to 316.97164917 Degrees (V3 158.0 to 178.0)											

Proposal 6362 - Observation 3 - Breaking the degeneracy: substellar anchors for evolutionary models

Observation	Proposal 6362, Observation 3: GJ758 Tue Nov 25 17:00:27 GMT 2025											
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(13)	GJ-758-B	RA: 19 23 33.9327 (290.8913862d) Dec: +33 13 18.31 (33.22175d) Equinox: J2000			Proper Motion RA: 81.966 mas/yr Proper Motion Dec: 160.158 mas/yr Epoch of Position: 2000						
Template	<i>Comments:</i> <i>Category=Star</i> <i>Description=[Brown dwarfs, Exoplanet Systems]</i> <i>Extended=NO</i>											
	TA Method					HFF Readout Mode						
	NONE					false						
Dithers	#	Dither Type		Size	Starting Point		Number of Points		Points			
	1	CYCLING		SMALL	1		9					
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	4	19	false	true	NONE	9	171	12473.501	177465
Special Requirements	Aperture PA Range 328.97164917 to 348.97164917 Degrees (V3 190.0 to 210.0)											

Proposal 6362 - Observation 4 - Breaking the degeneracy: substellar anchors for evolutionary models

Observation	Proposal 6362, Observation 4: HD33632											Tue Nov 25 17:00:27 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(4)	HD-33632	RA: 05 13 17.4478 (78.3226992d)				Proper Motion RA: -144.922 mas/yr					
			Dec: +37 20 14.32 (37.33731d)				Proper Motion Dec: -136.77200008714863 mas/yr					
			Equinox: J2000				Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[F stars] Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	5	5	false	true	NONE	9	45	3939.0	177465
Special Requirements	Aperture PA Range 15.97164917 to 35.97164917 Degrees (V3 237.0 to 257.0)											

Proposal 6362 - Observation 5 - Breaking the degeneracy: substellar anchors for evolutionary models

Observation	Proposal 6362, Observation 5: HD47127											Tue Nov 25 17:00:27 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(11)	HD-47127-B		RA: 06 37 33.9201 (99.3913338d)			Proper Motion RA: -74.864 mas/yr					
				Dec: +12 10 51.88 (12.18108d)			Proper Motion Dec: -283.98199995081086 mas/yr					
				Equinox: J2000			Epoch of Position: 2000					
	Comments: Category=Star Description=[Brown dwarfs, Exoplanet Systems] Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	5	4	false	true	NONE	9	36	3151.2	177465
Special Requirements	Aperture PA Range 208.97164917 to 228.97164917 Degrees (V3 70.0 to 90.0)											

Proposal 6362 - Observation 6 - Breaking the degeneracy: substellar anchors for evolutionary models

Observation	Proposal 6362, Observation 6: HIP21152											Tue Nov 25 17:00:27 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(6)	HIP-21152	RA: 04 32 4.8086 (68.0200358d)				Proper Motion RA: 112.174 mas/yr					
			Dec: +05 24 36.15 (5.41004d)				Proper Motion Dec: 7.756 mas/yr					
			Equinox: J2000				Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[F stars] Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	5	12	false	true	NONE	9	108	9453.601	177465
Special Requirements	Aperture PA Range 58.97164917 to 68.97164917 Degrees (V3 280.0 to 290.0)											

Proposal 6362 - Observation 7 - Breaking the degeneracy: substellar anchors for evolutionary models

Observation	Proposal 6362, Observation 7: HD72946											Tue Nov 25 17:00:27 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(7)	HD-72946	RA: 08 35 51.2667 (128.9636113d)				Proper Motion RA: -136.593 mas/yr					
			Dec: +06 37 21.95 (6.62276d)				Proper Motion Dec: -137.14800006709993 mas/yr					
			Equinox: J2000				Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[G stars] Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	5	2	false	true	NONE	9	18	1575.6	177465
Special Requirements	Aperture PA Range 223.97164917 to 243.97164917 Degrees (V3 85.0 to 105.0) Offset 0.153 arcsec, -0.107 arcsec											

Proposal 6362 - Observation 8 - Breaking the degeneracy: substellar anchors for evolutionary models

Observation	Proposal 6362, Observation 8: HR7672											Tue Nov 25 17:00:27 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(12)	HR-7672-B		RA: 20 04 6.1899 (301.0257912d)			Proper Motion RA: -387.472 mas/yr					
				Dec: +17 04 13.40 (17.07039d)			Proper Motion Dec: -419.4970000298781 mas/yr					
				Equinox: J2000			Epoch of Position: 2000					
	Comments: Category=Star Description=[Brown dwarfs, Exoplanet Systems] Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	5	2	false	true	NONE	9	18	1575.6	177465
Special Requirements	Aperture PA Range 358.97164917 to 18.97164917 Degrees (V3 220.0 to 240.0)											

Proposal 6362 - Observation 9 - Breaking the degeneracy: substellar anchors for evolutionary models

Observation	Proposal 6362, Observation 9: HD4747											Tue Nov 25 17:00:27 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(9)	HD-4747	RA: 00 49 26.7605 (12.3615021d)				Proper Motion RA: 519.049 mas/yr					
			Dec: -23 12 44.87 (-23.21246d)				Proper Motion Dec: 124.0409999999998 mas/yr					
			Equinox: J2000				Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[G stars] Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	5	6	false	true	NONE	9	54	4726.8	177465
Special Requirements	Aperture PA Range 21.97164917 to 41.97164917 Degrees (V3 243.0 to 263.0)											

Proposal 6362 - Observation 10 - Breaking the degeneracy: substellar anchors for evolutionary models

Observation	Proposal 6362, Observation 10: HD984											Tue Nov 25 17:00:27 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(10)	HD-984	RA: 00 14 10.2528 (3.5427200d)				Proper Motion RA: 104.775 mas/yr					
			Dec: -07 11 56.81 (-7.19911d)				Proper Motion Dec: -68.0160000001706 mas/yr					
			Equinox: J2000				Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[F stars] Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	5	6	false	true	NONE	9	54	4726.8	177465
Special Requirements	Aperture PA Range 188.97164917 to 197.97164917 Degrees (V3 50.0 to 59.0) Offset 0.124 arcsec, 0.217 arcsec											

Proposal 6362 - Observation 11 - Breaking the degeneracy: substellar anchors for evolutionary models

Observation	Proposal 6362, Observation 11: HIP-99770-2											Tue Nov 25 17:00:27 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(2)	HIP-99770	RA: 20 14 32.0324 (303.6334683d)				Proper Motion RA: 68.09 mas/yr					
			Dec: +36 48 22.70 (36.80631d)				Proper Motion Dec: 69.404 mas/yr					
			Equinox: J2000				Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[A stars] Extended=NO											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		9				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G395H/F290LP	NRSIRS2RAPID	5	6	false	true	NONE	9	54	4726.8	
Special Requirements	Aperture PA Range 236.97164917 to 246.97164917 Degrees (V3 98.0 to 108.0)											